Full Stack Development with MERN

Project Documentation Format

Smart Library Request Workflow

ServiceNow ITSM Implementation Guide

1. Introduction

- Project Title: Smart Library Request Workflow – ServiceNow ITSM
- Team Members: See team roles listed in the table below.

S.No.	Name	Role	Responsibility
1	Library Admin	System Owner	Workflow governance & SLA oversight
2	ServiceNow Developer	Technical Lead	Catalog, flow, and integration config
3	Circulation Staff	Fulfillment Agent	Physical resource handling & returns
4	Digital Resources Team	Fulfillment Agent	e-Book / e-Resource provisioning
5	End Users (Students / Faculty)	Requestors	Submit and track library requests

2. Project Overview

- Purpose: Digitalise all library resource requests through a unified ServiceNow self-service portal, automating assignment, SLA tracking, approvals, and notifications to improve patron experience and operational efficiency.
- Features:
  - Self-service portal for 7 request types (books, journals, e-resources, ILL, equipment, research help, room booking).
  - Automated triage, availability check via LMS REST API, and multi-level approvals.
  - Real-time SLA monitoring, escalation engine, and CSAT feedback loop.
  - RFID/barcode integration for physical pickup and return confirmation.

3. Architecture

Frontend (Service Portal)

- Built on ServiceNow Service Portal (AngularJS-based widgets).
- Catalog items per request type with dynamic intake forms.
- Responsive UI accessible on desktop and mobile browsers.

Backend (ServiceNow Platform)

- Flow Designer workflows orchestrate each request stage.

- Business Rules enforce state transitions and SLA triggers.
- Script Includes handle LMS REST API integration logic.

Database / CMDB

- Request records stored in ServiceNow Request [sc_request] and Request Item [sc_req_item] tables.
- LMS catalog synced via scheduled REST import to ServiceNow CMDB.
- Audit trail maintained in ServiceNow Activity log for compliance.

4. Setup Instructions

Prerequisites

- ServiceNow instance (Rome release or later).
- Library Management System (LMS) with REST API support (e.g., Koha, ALMA).
- LDAP / SSO configured for user authentication.
- SMS Gateway integration (e.g., Twilio) for outbound notifications.

Installation

- Step 1: Import the Smart Library Update Set into the ServiceNow instance.
- Step 2: Configure LMS REST API credentials in ServiceNow Connection & Credential Aliases.
- Step 3: Activate the Smart Library Service Catalog and assign catalog admin roles.
- Step 4: Set SLA definitions per catalog item category in SLA module.
- Step 5: Configure notification templates (email + SMS) for each workflow stage.
- Step 6: Test end-to-end flow using test user accounts in a sub-production instance.

5. Folder Structure

ServiceNow Application Files

- Update Sets: Smart_Library_v1.0.xml – full workflow, catalog, SLA, notification config.
- Flow Designer: Flows/SmartLibrary/ – one flow per request type.
- Script Includes: LMSApiHelper.js, LibraryNotificationUtils.js.
- Catalog Items: Service Catalog > Library Services > 7 catalog items.

Supporting Documents

- /docs/FSD – Smart_Library_ServiceNow_Workflow.docx (this document).
- /docs/API – LMS_REST_API_Spec.pdf.
- /docs/Test – Smart_Library_UAT_Scripts.xlsx.

6. Running the Application

- Access the ServiceNow Service Portal at: https://<instance>.service-now.com/sp
- Navigate to Library Services catalog category.
- Select the appropriate request type and complete the intake form.
  - For admin/fulfillment: log in with a Circulation Staff or Library Admin role.
  - For approvals: Librarians receive approval tasks in My Approvals module.

- Monitor all tickets via: Service Desk > Requests > Library Requests view.

7. Workflow Documentation

The table below documents all stages exposed by the Smart Library workflow, including request methods, responsible owners, and SLA targets.

S.No.	Stage	Description	Owner / Role	SLA
1	**Request Submission**	User logs into ServiceNow portal and submits a Smart Library request (book, journal, e-resource, equipment, or room).	End User	Immediate
2	**Auto-Acknowledgement**	ServiceNow auto-sends confirmation email/SMS with Request ID and expected fulfillment timeline.	System	Auto
3	**Triage & Assignment**	Library Admin reviews and assigns to Circulation, Acquisitions, or Digital Resources team.	Library Admin	2 Hours
4	**Availability Check**	REST API query to LMS/OPAC for real-time availability. Ticket status auto-updates.	System / Circulation	30 Mins
5	**Approval (if required)**	ILL or high-value requests routed through ServiceNow approval workflow to Librarian / Dept. Head.	Librarian / Dept. Head	1 Bus. Day
6	**Fulfillment**	Physical: shelf-pickup slot assigned. Digital: access link provisioned to user.	Circulation / Digital	4 Hours
7	**User Notification**	User receives pickup details or digital access credentials via email/SMS. Ticket: Pending User.	System	Auto
8	**Pickup / Access**	Barcode/RFID scan or digital log confirms delivery of physical or digital resource.	End User / Circulation	User-driven
9	**Ticket Closure**	Ticket auto-closes after confirmed pickup or 48-hour timeout. CSAT survey triggered.	System	48 Hours
10	**Return & Reconciliation**	Item scanned on return. Late return triggers automated fine notices. Digital access revoked at expiry.	Circulation / System	Per Policy

8. Authentication

- User authentication handled via institutional SSO (SAML 2.0 / LDAP).
- Role-based access control (RBAC) enforced in ServiceNow:
  - sn_sc_catalog: End users – submit requests only.
  - library_admin: Library Admins – triage, assign, monitor SLA.
  - library_approver: Librarians – approve ILL and high-value requests.
  - library_manager: Library Managers – view escalations and KPI dashboards.
- Session tokens managed by ServiceNow platform; no custom token logic required.

9. User Interface

Service Portal Catalog

- Home page: Library Services category with 7 request type tiles.
- Request form: Dynamic fields based on catalog item selection (e.g., ISBN search, date picker for room booking).
- My Requests page: Status tracker with stage badges (Submitted > In Progress > Ready for Pickup > Closed).

Request Categories & Catalog Items

S.No.	Request Type	Catalog Item (ServiceNow)	Priority	SLA
1	Physical Book Loan	LIB-BOOK-LOAN	Normal	Same Day
2	Journal / Periodical	LIB-JOURNAL-REQ	Normal	4 Hours
3	e-Book / e-Resource Access	LIB-EBOOK-ACCESS	High	1 Hour
4	Inter-Library Loan (ILL)	LIB-ILL-REQUEST	Normal	3–5 Bus. Days
5	Equipment Loan	LIB-EQUIP-LOAN	High	2 Hours
6	Research Assistance	LIB-RESEARCH-ASSIST	Normal	1 Bus. Day
7	Study Room Booking	LIB-ROOM-BOOK	Normal	Instant (Auto)

10. Testing

- Unit Testing: Script Include unit tests via ServiceNow ATF (Automated Test Framework).
- Integration Testing: LMS REST API mock tests using ServiceNow REST Message tester.
- UAT (User Acceptance Testing): Test scripts covering all 7 request types and 4 escalation scenarios.
- SLA Testing: Verify SLA timers trigger correctly at 50% and 100% elapsed.
- Notification Testing: Validate email and SMS delivery for all workflow stage triggers.

11. Screenshots or Demo

- Service Portal Home – Library Services tile view: [Insert screenshot]
- Book Loan Request Form: [Insert screenshot]
- My Requests – Status Tracker: [Insert screenshot]

- Library Admin – SLA Dashboard: [Insert screenshot]
- Approval Task – Librarian View: [Insert screenshot]
- Demo URL: https://<instance>.service-now.com/sp?id=library_demo

12. Known Issues

- LMS API timeout: If the LMS does not respond within 10 seconds, availability check defaults to 'Unknown' and manual verification is required. Workaround: Retry button on ticket.
- SMS delivery: SMS notifications may be delayed by up to 5 minutes during peak gateway load.
- ILL multi-level approval: Edge case where both approvers respond simultaneously may cause duplicate approval emails. Resolution in v1.1.

13. Future Enhancements

- AI-powered book recommendations: Suggest resources based on patron's borrowing history using ServiceNow Predictive Intelligence.
- Mobile push notifications: Native push via ServiceNow Mobile app for pickup alerts.
- QR code self-checkout: Patron-initiated checkout via QR scan, reducing circulation desk dependency.
- Analytics dashboard: Power BI / ServiceNow PA embedded dashboard for library KPI reporting.
- Chatbot integration: Virtual Agent chatbot on Service Portal for FAQ and quick request status lookups.

Appendix A – Escalation Matrix

S.No.	Trigger	Action	Escalated To	Notification
1	SLA > 50% elapsed	Warning alert to assigned staff	Assigned Staff	Email + In-App
2	SLA fully breached	Auto-escalate to Library Manager	Library Manager	Email + SMS
3	Approval overdue 1 business day	Reminder; auto-escalate at day 2	Dept. Head / CIO	Email
4	Item not returned by due date	Fine notice; 3-strike policy	User + Library Admin	Email + SMS

Appendix B – KPIs & Reporting

Tracked via ServiceNow Performance Analytics dashboards and reviewed at defined intervals.

S.No.	KPI	Target	Frequency
1	Request fulfillment rate	> 95%	Weekly
2	Average fulfillment time	< 4 Hours	Weekly
3	SLA compliance rate	> 98%	Monthly
4	CSAT (Customer Satisfaction) score	> 4.2 / 5.0	Monthly

5	Overdue return rate	< 5%	Monthly

This document is maintained by the Library IT Services team. For queries, raise a ticket under LIB-ADMIN on the ServiceNow portal.